BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Linear Search

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why must a trace table follow executed assignments in order?

2. What does lossy compression remove?

A) Only the file's name
B) All binary digits
C) Only duplicate file copies
D) Some original information

3. "Changing a variable name automatically fixes every logic error." Correct this. Explain why.

4. A program is compressed before being sent. "Lossless compression removes source-code characters." Correct this. Why does it matter here?

Today's objectives

Explain the steps and required conditions for linear search. Apply the algorithm or method to supplied data.

Key vocabulary - Read, cover, recall

Linear search
Checks one element after another.

Target
Value being searched for.

Index
Zero-based position of an array element.

Comparison
Test of current element against target.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Works without sorting	Linear search inspects items in sequence, so the list does not need to be sorted.
Check each element	Compare the target with the item at index 0, then increase the index if there is no match.
Report the result	Stop on the first match and return/report its index, or report not found after the last item.
Count comparisons	For n items, a search may check up to n items; finding an early match can require fewer.

I Do - One worked example

Trace linear search for target 14 in [6,12,9,14,20,3,11]. Use zero-based indexes. [4 marks]

1. At index 0

2. 6 · 14, so continue to the next element.

3. Index 1 = 12 and index 2 = 9 are not the target.

4. Index 3 = 14; found at index 3 after four comparisons. [4]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Find 15 in [4,8,12,15,19]. Give its index and comparisons. [4]

2. Find 7 in [9,3,8,2]. Describe the outcome. [3]

3. Explain why a linear search works on an unsorted list. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison that searches for the PC register.

```
array registers = ["MAR","MDR","PC","ACC"]
for i = 0 to 3
if ... then
print(i)
endif
next i
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. A linear search looks for 11 in [4,11,6,18,2]. State the index and number of comparisons. [3]

Q3. Explain why linear search can find a target in an unsorted list. [2]

Q4. A linear search fails to find 5 in [1,4,8,9]. How many elements are checked and why? [3]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Binary search can safely discard half only when data is sorted. ☐ T ☐ F
2. Linear search inspects items in sequence, so the list does not need to be sorted. ☐ T ☐ F
3. Compare the target with the item at index 0, then increase the index if there is no match. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Each variable value depends on the state produced by earlier executed instructions. ☐

Q2 D - Some original information ☐

Q3 Clear names show what values mean and help people read and change code. ☐

Q4 Lossless compression restores every character exactly. Losing characters could break the program. A thumbnail can lose some detail; source code cannot. ☐

Code First - Model

```
array registers = ["MAR","MDR","PC","ACC"]
for i = 0 to 3
if registers[i] == "PC" then
print(i)
endif
next i
```

Output: 2

We Do - Check the model

1. Compare 4,8,12,15 [2]; target found at index 3 [1], after four checks [1]. ☐

2. All four values are checked [2]; target not found [1]. ☐

3. It examines every element in sequence [1], does not rely on ordering [1] and stops at a match or the end [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Index 1 [1], following checks of 4 then 11 [1], total 2 comparisons [1]. ☐

Q3 Checks elements in order [1] without requiring sorting [1]. ☐

Q4 Four elements [1], because each item is compared [1] before concluding target absent [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: